
EECS 16A Designing Information Devices and Systems I

Homework 11

Submission Format

Your homework submission should consist of **one** file.

- `hw11.pdf`: A single PDF file that contains all of your answers (any handwritten answers should be scanned).

Submit the file to the appropriate assignment on Gradescope.

1. Reading Assignment

For this homework, please review Notes 18.5-18.6, Notes 19.5-19.6, and Note 20 (Op-Amp Current Source and Circuit Design).

- (a) Summarize the 4-step design process outlined in the notes, and describe why each step is useful in practice.

2. Op-Amp in Negative Feedback

(Contributors: Adhyyan Narang, Ava Tan, Aviral Pandey, Deepshika Dhanasekar, Lam Nguyen, Panos Zarkos, Titan Yuan, Vijay Govindarajan, Urmita Sikder, Sunash Sharma)

In this question, we analyze op-amp circuits in which the op-amp has a finite gain. Consider the circuit shown in Figure 1. Note that this circuit is in negative feedback. This circuit is known as a non-inverting amplifier circuit.

In parts (a) - (e), we will assume the op-amp in this circuit is ideal (i.e., gain is infinite), and then we will consider the case of finite gain in parts (f) - (h). For all parts, assume $V_{SS} = -V_{DD}$.

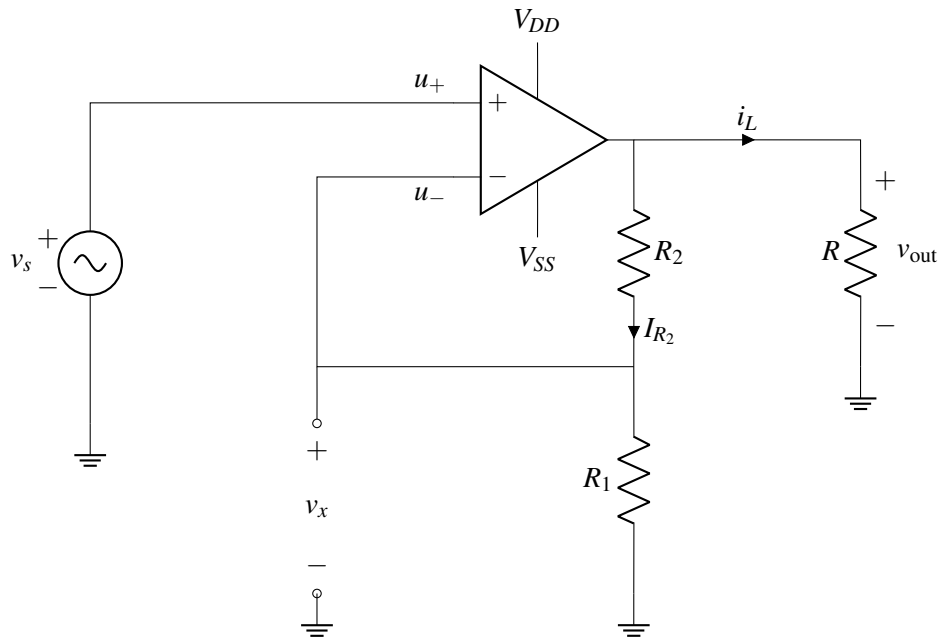


Figure 1: Non-inverting amplifier circuit

- Consider the circuit shown in Figure 1. What is $u_+ - u_-$?
- Find v_x as a function of v_{out} .
- What is I_{R_2} , i.e. the current flowing through R_2 as a function of v_s ? *Hint: Find the current through R_1 first.*
- Find v_{out} as a function of v_s .
- What is the current i_L through the load resistor R ? Give your answer in terms of v_{out} .
- We will now examine what happens when the op-amp gain is not ∞ . To understand what happens in this case, we will consider the equivalent circuit for an op-amp with gain A , shown in Figure 2. We will use this equivalent circuit to analyze the non-inverting amplifier circuit in Figure 1.

First draw an equivalent circuit for Figure 1, **by replacing the ideal op-amp in the non-inverting amplifier in Figure 1 with the op-amp model shown in Figure 2**. You do not need to include V_{DD} and V_{SS} in your equivalent circuit.

Then, using this setup, calculate v_{out} and v_x in terms of A , v_s , R_1 , R_2 and R . Assuming A is finite, is the magnitude of v_x larger or smaller than the magnitude of v_s ? Do these values depend on R ? *Hint: Note that the first golden rule still applies, i.e. the currents through the input terminals are zero.*

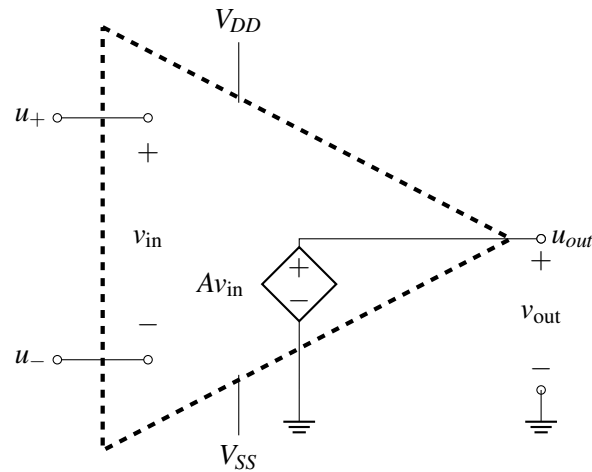


Figure 2: Internal op-amp model

- (g) Using your solution to the previous part, calculate the limits of v_{out} and v_x as $A \rightarrow \infty$. You should get the same answer as in part (d) for v_{out} .
- (h) **[OPTIONAL, CHALLENGE]** Now you want to make a non-inverting amplifier circuit whose gain is nominally $G_{nom} = \frac{v_{out}}{v_s} = 1 + \frac{R_2}{R_1} = 4$. However, G_{nom} can only be achieved only if the op-amp is ideal, i.e, if its internal gain $A \rightarrow \infty$. But, as with most considerations in the physical world, we must account for nonidealities! In reality, because you will be working with an op-amp with finite gain A , your designed circuit gain may come close to but will never quite reach G_{nom} as a result of the real op-amp's finite internal gain A .

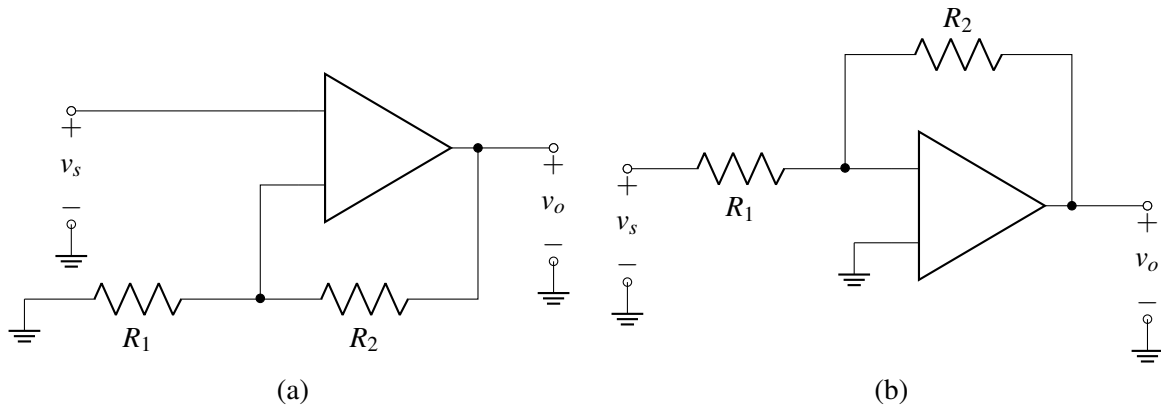
Suppose you would like your real op-amp circuit to have a minimum error of 1% (i.e, a minimum circuit gain of 3.96, i.e. $\frac{v_{out}}{v_s} \geq 3.96$). Remember that only if your op-amp were ideal, you would have a nominal circuit gain of $G_{nom} = \frac{v_{out}}{v_s} = 1 + \frac{R_2}{R_1} = 4$.

What is the minimum required value of A , called A_{min} , to achieve that specification? *Hint: Use your expression of v_{out} in part (f) to find an expression for $G_{min} = \frac{v_{out}}{v_s}$ when $A \neq \infty$.*

3. Basic Amplifier Building Blocks

(Contributors: Ava Tan, Aviral Pandey, Lam Nguyen, Michael Kellman, Titan Yuan, Vijay Govindarajan, Urmita Sikder)

The following amplifier stages are used often in many circuits and are well known as (a) the non-inverting amplifier and (b) the inverting amplifier.



- (a) Label the input terminals of the op-amp with (+) and (−) signs in Figure (a), so that it is in negative feedback. Then derive the voltage gain ($G = \frac{v_o}{v_s}$) of the non-inverting amplifier in Figure (a) using the Golden Rules. Why do you think this circuit is called a non-inverting amplifier?
- (b) Label the input terminals of the op-amp with (+) and (−) signs in Figure (b), so that it is in negative feedback. Then derive the voltage gain ($G = \frac{v_o}{v_s}$) of the inverting amplifier using the Golden Rules. Can you explain why this circuit is called an inverting amplifier?

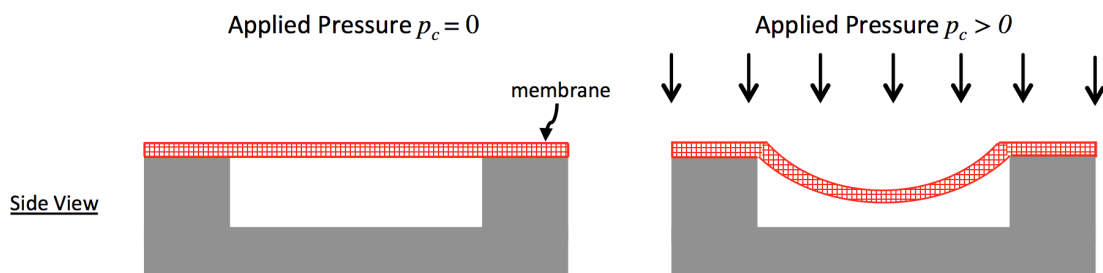
4. Putting on the Pressure: Build Your Own InstantPot

(Contributors: Craig Schindler, Ava Tan, Wahid Rahman, Urmita Sikder)

Prof. Ranade had a great experience with her automatic pressure cooker, so she was inspired to try and build her own. She's enlisting your help! The design of the pressure cooker uses a pressure sensor and a heating element. Whenever the pressure is below a set target value, an electronic circuit turns on the heating element.

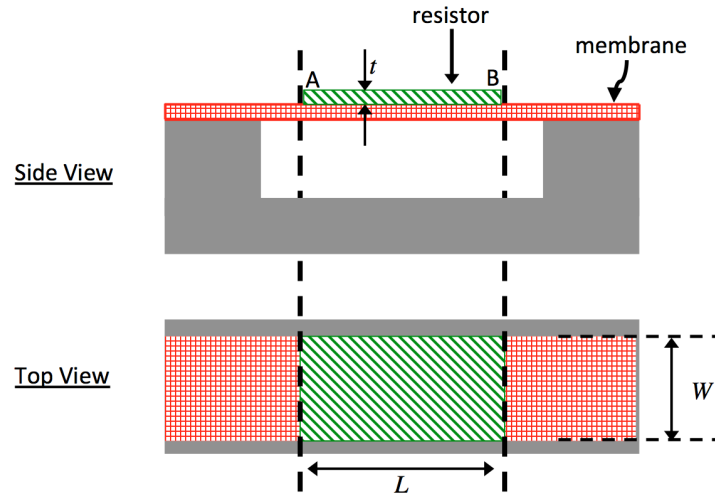
Pressure Sensor Resistance

The first step is designing a pressure sensor. The figure below shows your design. As pressure p_c is applied, the flexible membrane stretches.



- (a) You attach a resistor layer R_p with resistivity $\rho = 0.1\Omega\text{m}$, width W , length L , and thickness t to the pressure sensor membrane, as illustrated in the figure below. When the pressure $p_c = 0\text{Pa}$ (i.e. there is no applied pressure), $W = 1\text{mm}$, $L = L_0 = 1\text{cm}$, $t = 100\mu\text{m} = 100 \times 10^{-6}\text{m}$.

R_{p0} is the value of R_p when there is no applied pressure. Calculate R_{p0} . Note that direction of current flow in the resistor is from A to B as marked in the diagram.



- (b) When pressure is applied, the length of the resistor L changes from L_0 and is a function of applied pressure p_c , and is given by

$$L = L_0 + \beta p_c,$$

where L_0 is the nominal length of the resistor with no pressure applied, and β is a constant with units m/Pa. As a result of the length change, the value of resistance R_p also changes from its nominal value R_{p0} (the value of R_p with no pressure applied).

Derive an expression for R_p as a function of resistivity ρ , width W , thickness t , nominal length L_0 , constant β , and applied pressure p_c , when pressure is applied.

Note: The width and thickness of the resistor will also change with applied pressure. However, we ignore this to keep the math simple.

(c) **Pressure Sensor Circuit Design**

For this sub-part and the following sub-parts, we will use a new model for pressure-sensitive resistance R_p . Assume that the resistance R_p is a function of applied pressure p_c according to the relationship $R_p = R_o \times \frac{p_c}{p_{\text{ref}}}$, where $R_o = 1\text{k}\Omega$, and $p_{\text{ref}} = 100\text{kPa}$.

To complete our sensor circuit, we would like to generate a voltage V_p that is a function of the pressure p_c .

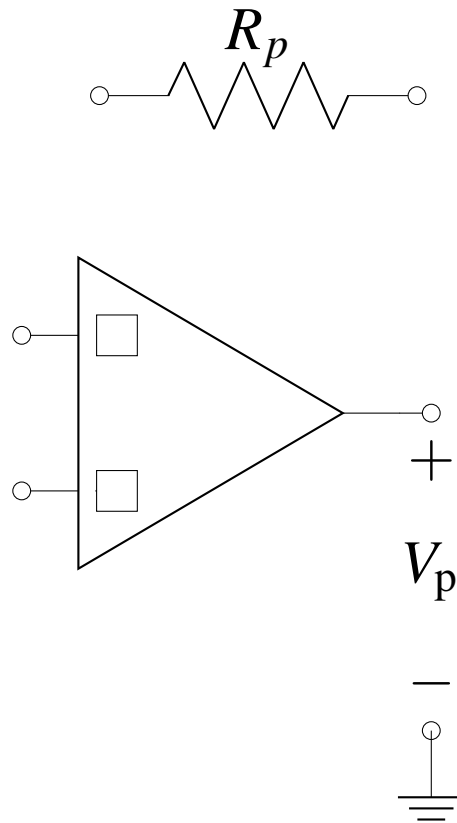
Complete the circuit below so that the output voltage V_p depends on the pressure p_c as:

$$V_p = -V_o \times \frac{p_c}{p_{\text{ref}}}, \text{ where } V_o = 1 \text{ V}.$$

Restrictions on your pressure sensor circuit design are as follows:

- **You may add at most one ideal voltage source and one additional resistor to the circuit, but you must calculate their values and mark them in the diagram.**
- **Mark the positive and negative inputs of the operational amplifier with “+” and “-” symbols, respectively, in the boxes provided.**
- **Assume op-amp supply voltages V_{DD} and $V_{SS} = -V_{DD}$ are already provided.**

You may assume that the operational amplifier is ideal.



(d) **Resistive Heating Element**

To heat the pressure cooker, you use a heating element with resistance R_{heat} . Calculate the value of R_{heat} such that the power dissipated is $P_{\text{heat}} = 1000 \text{ W}$ with $V_{\text{heat}} = 100 \text{ V}$ applied across the heating element.

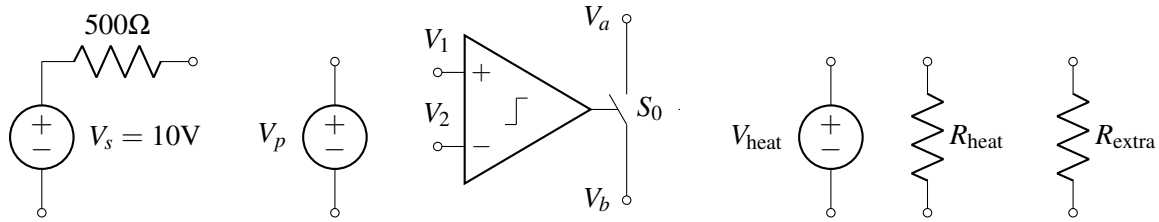
(e) **Pressure Regulation**

You are finally ready to complete the design of your pressure cooker.

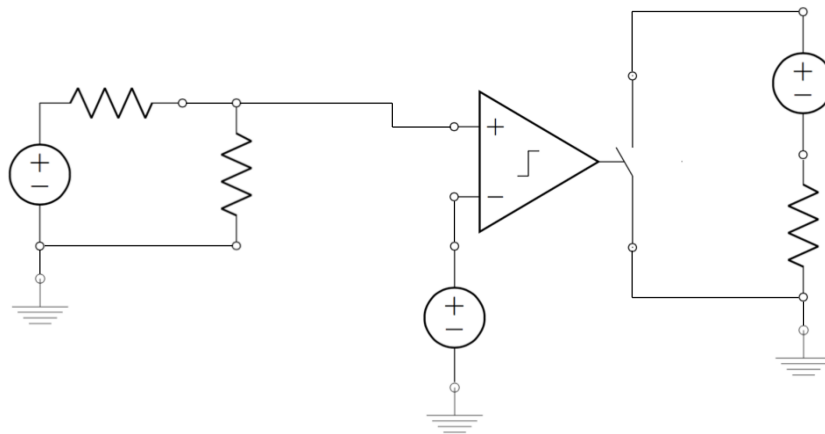
Using all of the circuit elements below, make a circuit that will turn the heater on (i.e. will cause a current to flow through R_{heat}) when the pressure is less than 500 kPa, and off (i.e. will cause no current to flow through R_{heat}) when the pressure is greater than 500 kPa.

The elements are:

- A voltage source $V_s = 10 \text{ V}$ in series with a resistance of 500Ω .
- A voltage source $V_p = V_o \times \frac{p_c}{p_{\text{ref}}}$, with $V_o = 1 \text{ V}$ and $p_{\text{ref}} = 100 \text{ kPa}$. (This is a voltage source whose voltage is a function of pressure p_c , unrelated to any previous parts of the question.)
- A comparator that controls switch S_0 . The switch is normally opened (i.e. an open circuit between nodes V_a and V_b), and is closed only when $V_1 > V_2$ (i.e. a short circuit between nodes V_a and V_b).
- The heater supply ($V_{\text{heat}} = 100 \text{ V}$).
- The heater resistor R_{heat} .
- One additional resistor R_{extra} that can have **any value**.
- You may assume you have access to a ground node.
- Assume comparator supply voltages V_{DD} and $V_{SS} = -V_{DD}$ are already provided.



- (i) Since you are looking to *compare* the change in voltage associated with a change in pressure, you decide to assign the **variable voltage** source V_p as one of the inputs to your comparator. What is the value of the **variable voltage source** V_p for $p_c = 500\text{kPa}$?
- (ii) For your comparator inputs, you also need to generate a **reference voltage** which can be used to compare against the value of the **variable voltage** source which you calculated above. Combine the voltage source $V_s = 10\text{V}$ with an associated resistance of 500Ω with an additional resistor R_{extra} to **generate a reference voltage equal to the voltage V_p you calculated above**. What would you choose the value of R_{extra} to be?
- (iii) Label the circuit elements in the schematic below with the **circuit elements presented above and your calculated R_{extra} value** to turn the heater on when the pressure is less than 500 kPa.



5. Island Karaoke Machine

(Contributors: Aviral Pandey, Lam Nguyen, Pangiotis Zarkos, Sashank Krishnamurthy, Titan Yuan, Urmita Sikder, Vijay Govindarajan)

Learning Goal: The objective of this problem is design a circuit that calculates the difference between two signals and amplifies the result.

You're stuck on a desert island and everyone is bored out of their minds. Fortunately, you have your EECS16A lab kit with op-amps, wires, resistors, and your handy breadboard. You decide to build a karaoke machine. You recover one speaker from the crash remains and use your iPhone as your source. You know that many songs put instruments on either the "left" or the "right" channel, but the vocals are usually present on both channels with equal strength.

In our case, the vocals are present on both left and right channels, but the instruments are only present on the right channel, i.e.

$$v_{\text{left}} = v_{\text{vocals}}$$

$$v_{\text{right}} = v_{\text{vocals}} + v_{\text{instrument}}$$

where the voltage source v_{vocals} can have values anywhere in the range of $\pm 120\text{mV}$ and $v_{\text{instrument}}$ can have values anywhere in the range of $\pm 50\text{mV}$.

What is the goal of a karaoke machine? **The ultimate goal is to *remove* the vocals from the audio output.** We're going to do this by first building a circuit that takes the left and right audio outputs of the smartphone and then calculates its **difference**. Let's see what happens.

The equivalent circuit model of the iPhone audio jack and speaker is shown in Figure 3. We model the **audio signals and jack** as v_{left} and v_{right} with **equivalent source resistance** of the left/right audio channels of $R_{\text{left}} = R_{\text{right}} = 3\Omega$. The **speaker** has an equivalent resistance of $R_{\text{speaker}} = 4\Omega$.

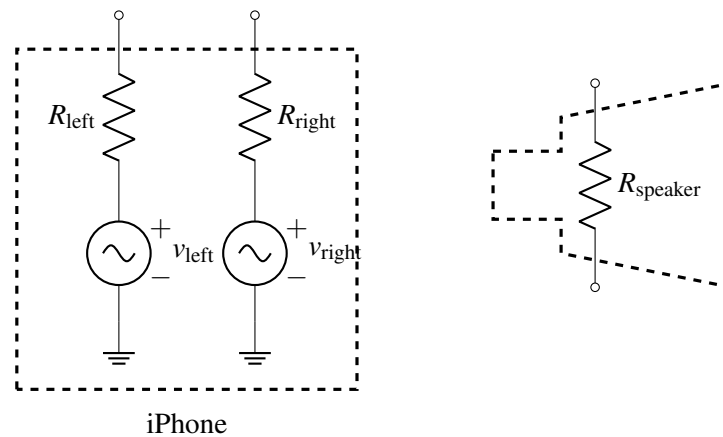


Figure 3: Audio jack and speaker of an iPhone.

- (a) One of your island survivors suggests the circuit in Figure 4 to do this. **Find the expression for the voltage across the speaker R_{speaker} as a function of v_{vocals} and $v_{\text{instruments}}$.** Does the voltage across the speaker depend on v_{vocals} ? In other words, what do you think the islanders will hear – vocals, instruments, or both?

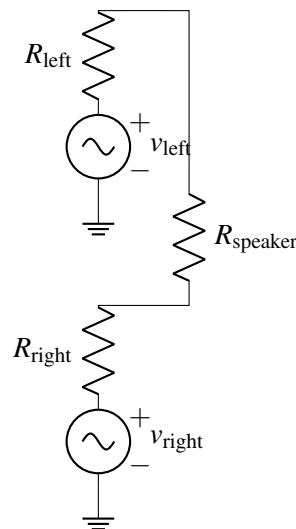


Figure 4: Circuit for part (a).

- (b) We need to boost the sound level to get the party going. To this end, we want a range of $\pm 2\text{ V}$ across the speaker. Design a circuit by completing the Figure 5 below that takes in $\{v_{\text{left}}, R_{\text{left}}\}$ and $\{v_{\text{right}}, R_{\text{right}}\}$ combos as inputs and outputs an **amplified version of $v_{\text{instrument}}$ across R_{speaker}** . Consider all op-amps to be **ideal** for this problem.

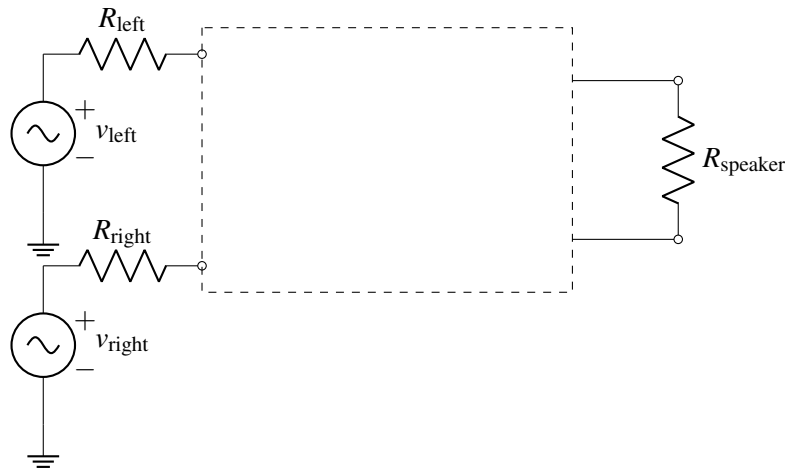


Figure 5: Circuit for part (b).

Hint 1: We need to get an output voltage with the range of $\pm 2\text{ V}$. The input voltage $v_{\text{instrument}}$ can have values anywhere in the range of $\pm 50\text{ mV}$. What gain is needed from the op-amp based amplifier circuits?

Hint 2: Use two op-amps in the non-inverting configuration. The non-ideal voltage source $\{v_{\text{left}}, R_{\text{left}}\}$ must be the input to one non-inverting amplifier and the non-ideal voltage source $\{v_{\text{right}}, R_{\text{right}}\}$ must be the input to the other non-inverting amplifier.

Hint 3: Connect the speaker R_{speaker} across the outputs of those two op-amps.

- (c) The trouble with the approach in part (b) is that two op-amps are required. Let's say you only have **one op-amp** with you. What would you do? One night in your dreams, you have an inspiration. Why not combine an inverting and non-inverting amplifier into one, as shown in Figure 6!

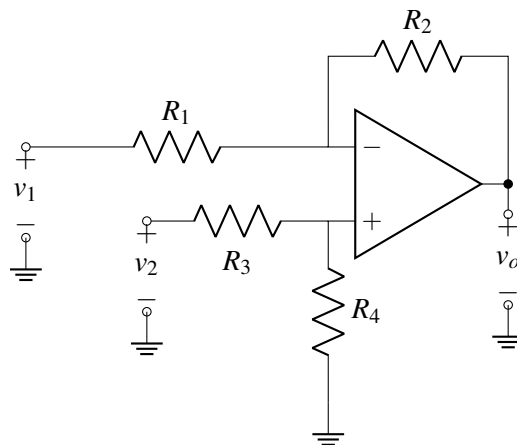


Figure 6: The new amplifier for part (c).

If we set $v_2 = 0\text{V}$, what is the output v_o in terms of v_1 ? (This is the inverting path.)

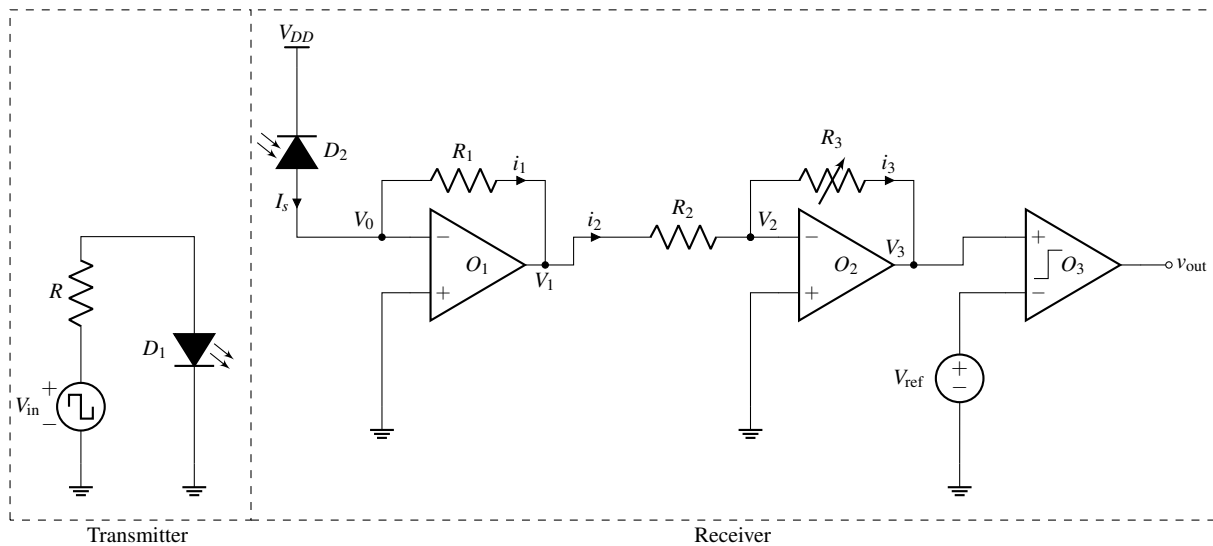
- (d) Consider the circuit in Figure 6 again. If we set $v_1 = 0\text{V}$, what is the output v_o in terms of v_2 ? (This is the non-inverting path.)
- (e) Now, determine v_o in terms of v_1 and v_2 by superposing your results from the last two parts. Choose values for R_1 , R_2 , R_3 and R_4 , such that the speaker has $v_o = \pm 2\text{V}$ across it for $v_2 - v_1 = \pm 50\text{mV}$.

6. Wireless Communication With An LED

(Contributors: Aviral Pandey, Lam Nguyen, Pangiotis Zarkos, Sashank Krishnamurthy, Titan Yuan, Urmita Sikder)

Learning Goal: This problem is designed to find the response of a complex circuit by breaking it into smaller circuit blocks and analyzing each block individually.

In this question, we are going to analyze the system shown in the figure below. It shows a circuit that can be used as a wireless communication system using visible light (or infrared, very similar to remote controls).



The element D_1 in the transmitter is a **light-emitting diode (LED)**. An LED is an element that emits light and whose brightness is controlled by the current flowing through it. You can recall controlling the light emitted by an LED in various labs during the course. In our circuit, the current through the LED, hence its brightness, can be controlled by choosing the applied voltage V_{in} and the value of the resistor R .

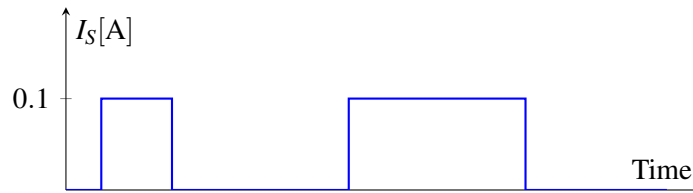
The light from the element D_1 in the transmitter is received by the element D_2 in the receiver. In this circuit, the LED D_1 is used as a means for **transmitting information with light**, and the element D_2 is used as a **light receiver to see if anything was transmitted**.

In the receiver, the element labeled as D_2 behaves like a **"reverse biased" solar cell**, which means that when it receives light, it generates a current. We will denote the current generated by the solar cell by I_s .

Remark: Non-idealities, such as background light, affect the performance of a system that does light measurements. In this question, we will assume ideal conditions, that is, there is no source of light around except for the LED.

In our system, we define two states for the transmitter: (i) the transmitter is sending something when **they turn on the LED** and (ii) the transmitter is not sending anything when **they turn off the LED**. On the receiver side, the goal is to **convert the current I_S generated by the solar cell into a voltage and amplify it**, so that we can read the output voltage V_{out} to see if the transmitter was sending something or not. The circuit implements this operation through a series of op-amps and a comparator. *It might look complicated at first glance, but we can analyze it one section at a time.*

- Currents i_1 , i_2 and i_3 are labeled on the diagram. Assuming *op-amps are ideal* and the Golden Rules hold, is $I_S = i_1$? $i_1 = i_2$? $i_2 = i_3$? Treat the solar cell as an ideal current source supplying I_S .
- Use the Golden Rules to find V_0 , V_1 , V_2 and V_3 in terms of I_S , R_1 , R_2 and R_3 .
Hint: Solve for V_0 first, then use V_0 to find V_1 . Afterwards, find V_2 and V_3 in a similar manner.
- Now, assume that the transmitter has chosen the values of V_{in} and R to control the intensity of light emitted by LED, such that when the **transmitter is sending something**, I_S is equal to 0.1 A and when the **transmitter is not sending anything**, I_S is equal to 0 A. The following figure shows a visual example of how this current I_S might look like as time changes.



For the receiver, suppose $V_{ref} = 2\text{ V}$, $R_1 = 10\ \Omega$, $R_2 = 1000\ \Omega$, and the supply voltages of the comparator are $V_{DD} = 5\text{ V}$ and $V_{SS} = -5\text{ V}$. Pick a value of R_3 such that V_{out} is V_{DD} when the transmitter is sending something (i.e. $I_S = 0.1\text{ A}$) and V_{SS} when the transmitter is not sending anything (i.e. $I_S = 0$).

7. Homework Process and Study Group

Who did you work with on this homework? List names and student ID's. (In case you met people at homework party or in office hours, you can also just describe the group.) How did you work on this homework? If you worked in your study group, explain what role each student played for the meetings this week.